Meta-Learning for 1D-ARC

Assignment 1 Report

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Problem Statement

The **1D-ARC benchmark** is a simplified version of the Abstraction and Reasoning Corpus (ARC), restricted to one-dimensional sequences of integer tokens (values 0–9, with 20 as a padding sentinel). The dataset contains **901 tasks** drawn from **18 concept classes**, each task providing 2–4 input/output demonstration pairs (the *support set*) and one query input whose output must be predicted.

Each task tests a specific transformation concept such as filling, flipping, mirroring, moving, scaling, or recolouring objects. The core challenge is to infer the transformation rule from only 2–4 examples and generalise it to a novel query — exactly the few-shot meta-learning problem.

Two evaluation regimes are studied:

- **Same-Concept (SC):** train and test tasks share the same 18 concepts; 80/20 stratified split by concept.
- **Cross-Concept (CC):** 14 concepts used for training, 4 held-out concepts used for testing.

Dataset and Setup

- **Total tasks:** 901 (18 concept classes, ≈50 tasks each). Tokens: integers 0–9; `PADDING_VALUE = 20`.
- **Same-concept split:** 720 train / 181 test. With 3× colour-swap augmentation: 2160 augmented training tasks.
- **Cross-concept split:** 701 train tasks (14 concepts) / 200 test tasks (4 held-out: `fill`, `hollow`, `move_2p`, `move_dp`).
- **Augmentation:** Colour-swap — non-zero, non-background token colours are randomly permuted while leaving background (0) positions unchanged.

Models

BiLSTM Baseline (Standard Supervised Learning)

A bidirectional LSTM processes each (input, output) sequence pair. All support pairs are pooled across tasks with no episodic structure. At test time the model receives the query input but has no mechanism to use the support set. This establishes a lower bound.

BiLSTM + FOMAML

The same BiLSTM architecture is trained with First-Order MAML: an inner loop runs SGD steps on the support set of each task, and the meta-gradient updates the shared initialisation. At test time, the model adapts using 20 SGD steps on the task's support pairs before predicting the query.

Matching Networks (Episodic)

Matching Networks use an attention mechanism over the support set: the query is compared to all support inputs via cosine similarity, and the predicted output is a weighted combination of support outputs. Training is episodic, directly matching the test-time structure.

PosCNP (Positional Conditional Neural Process)

PosCNP extends the Conditional Neural Process with a cross-attention module incorporating positional encodings. Support pairs are encoded into latent representations, aggregated, and decoded for the query via positional cross-attention. The positional component is well-suited to 1D-ARC since tasks like move and flip are inherently position-sensitive.

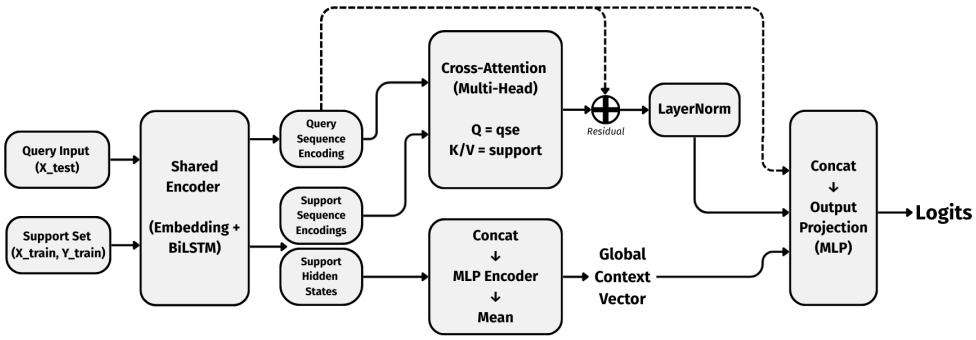

Figure 1: PosCNP architecture: support pairs are encoded and aggregated; the query is decoded via positional cross-attention.

Same-Concept Results

Table 1: Same-concept split test accuracy. Augmented = 3× colour-swap augmentation.

Model	Original	Augmented
BiLSTM (supervised)	30.9%	44.2%
BiLSTM + FOMAML	29.8%	61.3%
Matching Networks	55.8%	75.7%
PosCNP	**69.6%**	**78.5%**

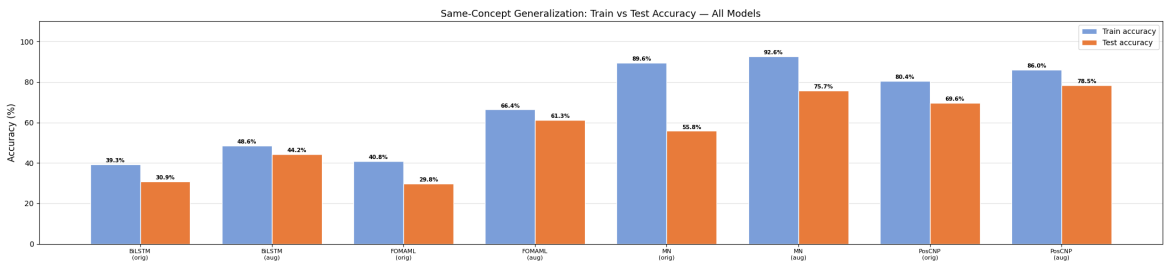

Figure 2: Train vs test accuracy for all models on the same-concept split.

Key observations:

- **Augmentation consistently helps.** FOMAML benefits the most (+31.5 pp), requiring diverse tasks for a good meta-initialisation.
- **Episodic models outperform supervised.** Matching Networks and PosCNP significantly outperform the BiLSTM baseline on original data.
- **PosCNP achieves 78.5%** (augmented) — the best same-concept result — due to its positional cross-attention suited to ARC transformations.
- **FOMAML without augmentation underperforms** the supervised baseline (29.8% vs 30.9%), as inner-loop adaptation overfits small support sets without diverse training data.

Per-Concept Breakdown

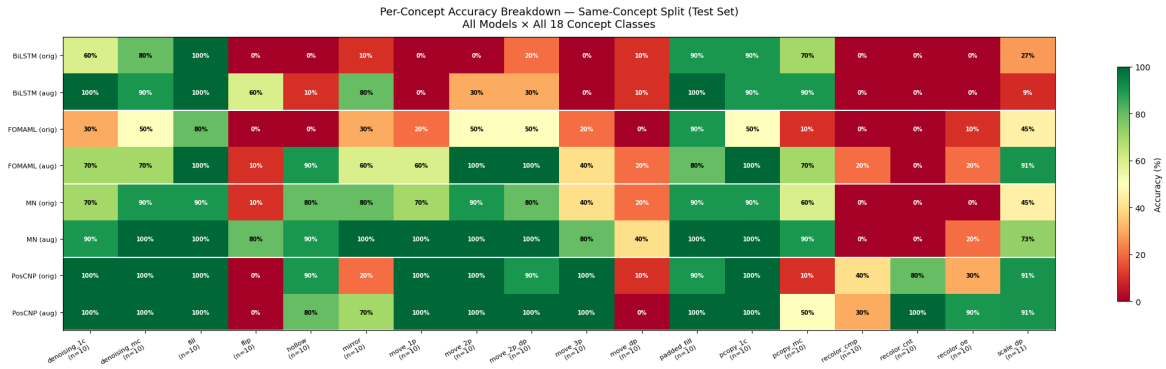

Figure 3: Per-concept accuracy heatmap (same-concept test split). Green = high, Red = low accuracy.

The heatmap reveals a clear pattern: **recolouring tasks are consistently hard across all models**. Concepts such as `recolor_cmp` and `recolor_cnt` require counting objects to determine colour assignment, which none of the architectures handle well. Simple positional tasks like `flip` and `mirror` are reliably solved at high accuracy, confirming that difficulty is intrinsic to the concept.

Cross-Concept Results

Table 2: Cross-concept split test accuracy. Held-out: `fill`, `hollow`, `move_2p`, `move_dp`.

Model	Original	Augmented
BiLSTM (supervised)	28.0%	24.5%
FOMAML (with adaptation)	43.0%	38.5%
PosCNP+MAML (with adaptation)	48.0%	**61.0%**

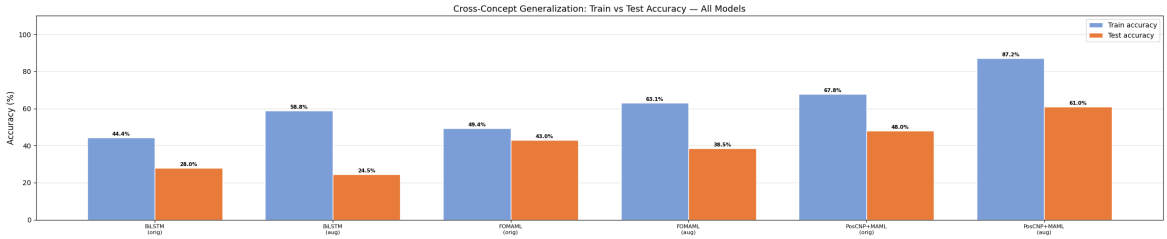

Figure 4: Train vs test accuracy for all models on the cross-concept split.

Key observations:

- **Adaptation is essential.** Removing test-time adaptation drops accuracy by 10–20 pp for both FOMAML and PosCNP+MAML.
- **Augmentation can hurt BiLSTM** on cross-concept (24.5% vs 28.0%) — more data without episodic structure causes overfitting to training concepts.
- **PosCNP+MAML (augmented) achieves 61%**, suggesting positional cross-attention generalises across concept boundaries.

Per-Concept Breakdown

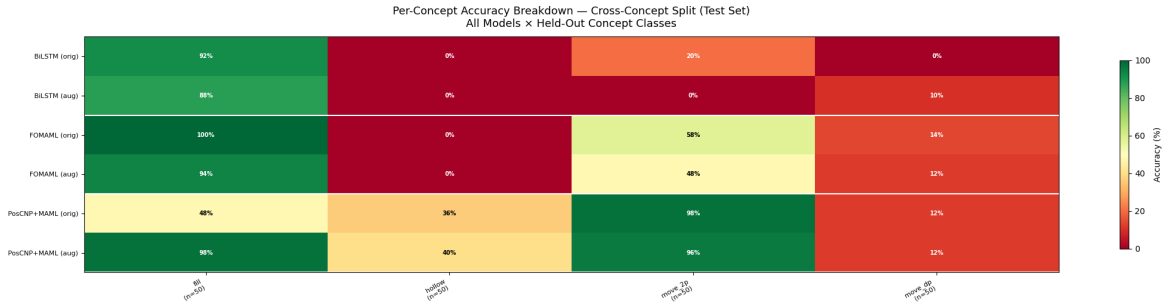

Figure 5: Per-concept accuracy for all models on the 4 held-out cross-concept classes.

The per-concept heatmap shows `fill` is solved at 88–100% by most models, while `hollow` scores 0% for BiLSTM and FOMAML but 36–40% for PosCNP+MAML. `move_dp` remains near 0% across all models. This highlights that the reported cross-concept accuracy is **highly split-dependent**.

Robustness: Random-Split Averaging

To assess whether the fixed split is representative, PosCNP+MAML was evaluated across 5 random splits, each holding out a different set of 4 concepts (5 epochs of training per round).

The high variance (10.1 pp) confirms that single-split cross-concept reporting can be misleading. When recolouring concepts are held out, accuracy falls to 3–8%; when simpler concepts are held out, it reaches 32%.

Table 3: Random-split averaging results (PosCNP+MAML, 5 rounds).

Round	Held-out Concepts	Accuracy
1	denoising_1c, flip, move_2p_dp, scale_dp	32.8%
2	denoising_1c, fill, move_3p, recolor_oe	17.0%
3	hollow, move_2p_dp, move_3p, padded_fill	15.5%
4	denoising_1c, denoising_mc, recolor_cmp, recolor_cnt	8.0%
5	mirror, padded_fill, pcopy_1c, recolor_cmp	3.0%
Mean ± Std		**15.3% ± 10.1%**

Analysis and Discussion

Why Augmentation Helps

Colour-swap augmentation triples the training set without introducing semantically new examples, forcing models to learn structural transformations rather than memorising colour patterns. The benefit is largest for FOMAML which requires many diverse tasks to learn a good meta-initialisation.

Why Recolouring Is Hard

Recolouring tasks require counting objects or comparing quantities before assigning colours. None of the architectures have an explicit counting mechanism. Future work should explore slot attention or object-centric encoders that can capture object-level structure and enable counting-based reasoning.

FOMAML: Fast Adaptation at a Cost

FOMAML gives an edge in cross-concept generalisation (43% with adaptation vs 28% BiLSTM supervised), but the raw initialisation without adaptation performs very poorly (16%), indicating the model relies heavily on test-time adaptation rather than a strong prior.

Conclusion

- **Best same-concept:** PosCNP (augmented) — **78.5%**
- **Best cross-concept:** PosCNP+MAML (augmented) — **61.0%**
- **Augmentation** is essential, especially for FOMAML (+31.5 pp same-concept).
- **Recolouring tasks** remain unsolved; explicit counting mechanisms are needed.
- **Cross-concept scores are split-dependent:** 15.3% ± 10.1% over random splits.
- **Positional cross-attention** (PosCNP) is the right inductive bias for position-sensitive ARC transformations.